CSE/W-24/CS3T001/AI3T001/02

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

Education to Eternity

॥कृन्वन् सर्वं सभ्यम्॥

B. Tech ~~Summer~~ / Winter – 2024

(ESE / ~~Make-up~~ / ~~Supplementary~~) Examination

Branch: Computer Science & Engineering/Artificial Intelligence Sem: III

Subject: Computer Network Subject Code: CS3T001/ AI3T001

Marks: 100 Date of Exam: 13/11/2024 Time: 3 Hrs.

Instructions to the students:

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Assume suitable data whenever necessary.**
4. **Illustrate your answer whenever necessary with the help of neat sketches.**
5. **Use of Programmable Calculator is prohibited.**
6. **Don't use red pen for writing the answers.**

COs: Students should be able:

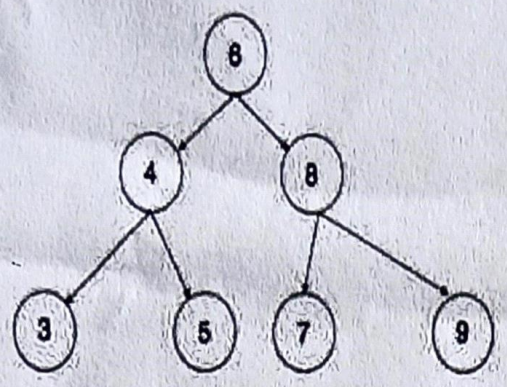
1. To understand the basics of network and its hardware components.
2. To classify the different layers of Network Models.
3. To analyse the services and algorithm of Network Models.
4. To categorize the different transmission media with its connectors.
5. To implement the network security and privacy in model.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one/two word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Topology with example.	L1/CO1	2
Q.2	State the Data Representation Methods in Computer Network.	L1/CO1	2
Q.3	Identify the types of Network.	L1/CO1	2
Q.4	List out the Elements of Transport Protocols.	L1/CO3	2
Q.5	Write short note on Transmission Impairment.	L3/CO2	2
Q.6	Classify the different Security Services in network.	L2/CO5	2
Q.7	Restate Definition of IP Address with example.	L2/CO2	2
Q.8	A signal has frequency 3125MHz, 254MHz, 9000MHz & 200 MHz Find the Bandwidth of signal.	L1/CO2	2
Q.9	Consider the following CRC generator and Find the corresponding binary pattern? a) $x^5 + x^4 + x^2 + 1$ b) $x^{10} + x^7 + x^4 + x^3 + x + 1$	L1/CO1	2

Q.10	Find the Shortest Path of give tree	L3/CO3	2



Q.11	Rewrite types of TCP Timers.	L2/CO2	2
Q.12	Enlist the Application Layer Services.	L1/CO5	2

SECTION B

1. **All Questions are compulsory.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Discuss the Retransmission with its types in details.	L2/CO4	4
Q.14	Illustrate the Network Layer Design Issues in details.	L3/CO2	4
Q.15	Calculate weather 192.1.1.47 to 192.1.1.60 IP address is valid IP address block or not.	L4/CO5	4
Q.16	We need to send 165 Kbps over a Noiseless Channel with bandwidth of 20KHz. How many signal levels do we need?	L4/CO4	4
Q.17	Write short note on WWW and HTTP.	L3/CO5	4
Q.18	Find the Classes of Given IP address.	L3/CO4	4

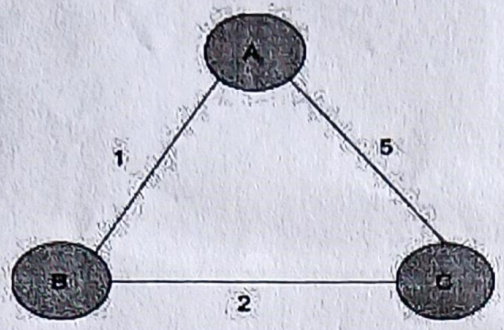
a) 251.10.2.27
b) 20.10.1.10
c) 11111101 00011101 10101010 11100101 10101010
d) 10111000 11100011 00010101 10110001 11001010

SECTION C

1. **Solve any FOUR Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Illustrate Data Communication with its Characteristics and Components in details.	L4/CO2	7
Q.20	Consider the message sender wants to send is 100100, and the generator polynomial is $x^3 + x^2 + 1$. Find the message transmitted by the sender. If the receiver receives the message, check if the receiver receives the correct message or not.	L5/CO4	7
Q.21	Explain with diagram Connection Establishment and Termination in TCP.	L5/CO4	7
Q.22	Consider the following network; three routers are there: A, B, and C Find the routing table for all the routers A, B, and C by using Distance Vector	L4/CO5	7

Routing.



Q.23	Describe Routing Algorithms with its types in details.	**L6/CO3**	**7**
Q.24	Discuss Computer Networks with its Advantages and Disadvantages	**L2/CO2**	**7**

SECTION D

1. **Solve any TWO Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Explain the following layer:(any 4) **a)** Physical Layer. **b)** Data Link Layer **c)** Network Layer **d)** Transport Layer **e)** Session Layer **f)** Presentation Layer	**L5/CO5**	**12**
Q.26	Differentiate between following **a)** Data **Vs** Signal **b)** PAN **Vs** LAN **Vs** MAN **Vs** WAN **c)** Connectionless **Vs** Connection Oriented Service	**L4/CO3**	**12**
Q.27	Describe OSI Reference Model & TCP/IP Reference Model with its layers in details.	**L6/CO4**	**12**